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Adequacy aware long-term energy-system optimization models considering stochastic peak demand

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ABSTRACT

An important aspect of long-term power system planning models is their adequacy awareness, i.e., their ability to ensure generation adequacy in the final solution, which in turn strongly depends on the level of temporal detail included in the model structure. To maintain computational tractability, the temporal detail included in long-term planning models is, however, often limited, which decreases their adequacy awareness resulting in inadequate capacity mixes. To compensate for this, several adequacy-improving measures can be taken. The aim of this paper is to investigate the performance of three of these measures, namely (i) adding a traditional static planning reserve margin (PRM) constraint, (ii) adding a dynamic PRM constraint and (iii) a novel approach that increases the level of temporal detail with which critical peak periods are modeled. To this end, we compare each of these three adequacy methods in a long-term planning exercise based on (i) total system costs, (ii) adequacy of the resulting capacity mix and (iii) technology choices. Our results suggest that the novel approach more effectively increases the adequacy of the obtained capacity mix, resulting in lower total system costs and less technology biases. Also the dynamic PRM approach performs well, although occasional inadequacies can occur in individual milestone years due to exogenously imposed capacity credits.

1. Introduction

Long-term energy-system optimization models (ESOMs), such as TIMES [1] are widely used to help analyze and guide the ongoing transitions in the energy system. These models often have a detailed bottom-up, techno-economic representation of the various energy sectors, and are therefore well suited to investigate the complex interactions between technologies and commodities. However, to maintain computational tractability, these models usually have a limitedly detailed temporal structure, i.e., they include only a few time slices representing the inter- and intra-annual variations in demand and availability of renewable energy sources. Obviously, a limitedly detailed temporal structure decreases the accuracy with which the characteristics of weather-dependent power flows from intermittent renewable energy sources (iRES) are captured [2,3]. For this reason, several tools have been developed that allow modelers to better capture these intermittent characteristics of iRES technologies with a limited amount of time slices, see e.g., [4,5].

In such ESOMs with a limited temporal structure, it may, however, remain challenging to ensure that the resulting capacity mix is adequate. Generation system adequacy is often loosely defined as the ability of a power system to provide an adequate supply of electricity in order to meet the demand under reasonable conditions [6]. Mertens et al. [7,8] illustrate that a simplified temporal representation provides the model typically with too little information about the dispatch situations during peak moments, resulting in an inaccurate view (i) on the peak demands and the capacity required to cover it, as well as (ii) the availability of novel resources, such as intermittent renewables and energy storage systems, during moments of peak demand (i.e., their capacity credits). Furthermore, as explained by Merrick [9] the inaccurate representation of these critical peak demand situations can also affect overall technology choices.

Not considering appropriate measures to ensure generation system adequacy in energy system planning may have several ramifications. First, this may distort or bias power system metrics such as the share of renewables or greenhouse gas emissions [7,8]. Second, acknowledging that these planning models are often used to support policy making, these distortions can strengthen or weaken conclusions from scenario analyses. Furthermore, comparisons between scenarios may be flawed if a consistent adequacy treatment is lacking since comparing systems

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Nomenclature

Y	Set of all milestone years, indexed by y.
V	Set of all vintage years, indexed by v.
I	Set of all representative days, indexed by i.
T	Set of all time steps in each representative day, indexed by t.
G	Set of all generation technologies, indexed by g
G_D	Set of all dispatchable generation technologies, $G_D \subset G$.
G_R	Set of all renewable generation technologies, $G_R \subset G$.
S	Set of all newly installed storage technologies, indexed by s.
P	Set of all critical peak periods, indexed by p.
$H(p)$	Set of all hours in critical peak period p, indexed by h.
$VOLL$	Value of lost load, [e/MWh].
DF_y	Discount factor for milestone year y, [-].
$N^{PY}_{g/s,y}$	Discounted number of period years in milestone year y, [-].
W_i	Weight of representative day i, [-].
W^{SAC}	Weight of each critical peak hour, [-].
Δ	Duration of each time step, [h].
$D_{y,i,t}$	Electric power demand at time step t of representative day i in milestone year y, [MW].
$D^{SAC}_{y,h,p}$	Electric power demand at hour h of peak period p in milestone year y, [MW].
$CF_{g,i,t}$	Capacity factor of renewable technology g at time step t of representative day i, [-].
$CC_{g/s}$	Capacity credit of generation technology g or storage technology s, [-].
$\eta_{s,v}$	Roundtrip efficiency of storage technology s installed in year v, [-].
PRM	Planning reserve margin, [-].
$MC_{\hat{g},y}$	Marginal cost of the peak generator $\hat{g}$ in milestone year y, [e/MWh].
c^{inv}_y	Total investment costs associated with milestone year y, [e].
c^{fom}_y	Total fixed operation & maintenance costs associated with milestone year y, [e].
c^{vom}_y	Total variable operation & maintenance costs associated with milestone year y, [e].
c^{gen}_y	Total generation costs associated with milestone year y, [e].
c^{ens}_y	Total load shedding costs associated with milestone year y, [e].
c^{dummy}_y	dummy generation cost associated with milestone year y, [e].
$cap^{av}_{g,y}$	Capacity of generation technology g installed in year v that is available in milestone year y, [MW].
$cap^{av,P}_{s,y}$	Power capacity of storage technology s installed in year v that is available in milestone year y, [MW].
$cap^{av,E}_{s,y}$	Energy capacity of storage technology s installed in year v that is available in milestone year y, [MWh].
$gen_{g,v,y,i,t}$	Electric power generation of technology g installed in year v at time step t of representative day i in year y, [MW].
$gen^{dummy}_{y,i,t}$	Dummy generation variable at time step t of representative day i in year y, [MW].
$ch_{s,v,y,i,t}$	Charge variable of technology s installed in year v at time step t of representative day i in year y, [MW].
$dc_{s,v,y,i,t}$	Discharge variable of technology s installed in year v at time step t of representative day i in year y, [MW].
$ens_{y,i,t}$	Load shedding at time step t of representative day i in year y, [MW].
$gen^{SAC}_{g,v,y,p,h}$	Electric power generation of renewable technology g installed in year v at hour h of peak period p in year y, [MW].
$ch^{SAC}_{s,v,y,p,h}$	Charge variable of technology s installed in year v at hour h of peak period p in year y, [MW].
$dc^{SAC}_{s,v,y,p,h}$	Discharge variable of technology s installed in year v at hour h of peak period p in year y, [MW].
$e^{SAC}_{s,v,y,p,h}$	Energy level of technology s installed in year v at hour h of peak period p in year y, [MWh].
$ens^{SAC}_{y,p,h}$	Load shedding at hour h of peak period p in year y, [MW].

with a different degree of adequacy is unfair. Finally, whenever one is a faced with an adequacy problem and one is interested in finding a cost-optimal pathway towards an adequate system, an adequacy aware planning model can serve as a first estimate. This first estimate can subsequently be used as a basis for further analysis to assess which additional measures need to be taken to ensure system adequacy.

The adequacy concern in ESOMs has traditionally been addressed by adding a constraint that imposes a lower limit on the firm capacity, i.e., the capacity that can be relied on during moments of peak demand. This constraint is often referred to as the planning reserve margin constraint, since it aims to mitigate the probability of involuntary load shedding by forcing the firm capacity to exceed the peak demand by a planning reserve margin (PRM). An important concept in this regard is the capacity credit, i.e., a metric that reflects the share of the nameplate capacity of a technology that can accounted for as firm capacity, especially for intermittent renewables [10] and energy storage systems [11]. Many ESOMs allow for the introduction of traditional PRM constraints (see for example TIMES [1] or OSeMOSYS [12]). However, no clear set of guidelines exists to determine the parameters involved, i.e., the planning reserve margin as well as the capacity credit for variuous technologies. As such, different designs of this PRM constraint are encountered in the literature. For example, the analysis performed by Welsh et al. [13] uses a PRM value of 27% and the capacity credits for wind are determined by the analytical formula proposed by Voorspools and D'haeseleer [14]. The MARKAL model by Anandarajah et al. [15] assigns capacity credits to several intermittent technologies depending on each technology's current penetration. Capacity credit values are inspired by model runs with a dedicated power-systems model [16]. The model used in [17] and [18] assumes intermittent wind and solar PV technologies do not make a capacity contribution, i.e., their capacity credit is assumed to be zero.

Recently, other, more elaborate methods to improve the adequacy awareness of long-term planning models have been used. Two main strategies can be identified in this regard, namely (i) using dynamic PRM constraints and (ii) improving the temporal detail with which peak periods are modeled [19]. Dynamic PRM constraints are PRM constraints of which the parameters involved are estimated by an external module and updated in between investment periods. Improving the temporal detail with which peak periods are modeled usually entails including a so-called peak time slice to expose the model to an extreme peak situations in which the firm capacity becomes scarce. We will discuss these two approaches in more detail in Section 2.

Several gaps in the current literature about the adequacy awareness of planning models can still be identified. First, the exact impact of the temporal detail on the adequacy awareness of a long-term planning model has to the best of the authors' knowledge not yet been analyzed. Second, a thorough comparison of the two strategies mentioned above has not yet been made. As such, potential pitfalls of these strategies that cause inaccurate model outcomes might be overlooked. Third, improving the temporal detail with which peak periods are modeled is often done by adding a few peak time slices in the temporal structure of the planning model. A particular challenge associated with planning models as is that the characteristics of these peak time slices (e.g., demand, iRES generation) depend on the underlying capacity mix, and since the capacity mix is not known before solving a planning model it is not straightforward to make an accurate selection of peak time slices a priori. This is especially the case in a planning model with a long panning horizon and hence, a comprehensive application of this strategy in a long-term dynamic planning model (as opposed to a static, single year planning model) has not yet been done.

Corresponding to these literature gaps, this paper aims to contribute to the existing literature as follows:

1. The impact of the temporal detail on the adequacy awareness of planning models is analyzed by comparing planning model outcomes from several limitedly detailed planning models with the outcomes of a highly detailed planning model.
2. A case study that compares different adequacy methods is conducted. In this case study, we analyze the performance of (i) a traditional PRM constraint, (ii) a dynamic PRM constraint and (iii) improving the temporal detail with which critical peak periods are modeled. Emerging from this case study is a set of guidelines for modelers that seek to improve the adequacy awareness of their planning models.
3. A novel model formulation to increase the temporal detail with which potential peak periods are modeled in a long-term dynamic planning model setting is proposed. This formulation features a set of adequacy constraints that are evaluated for a carefully selected set of peak periods. The novelty is that we decouple the temporal representation used to model the day to day iRES and load characteristics from the temporal representation used to model the critical peak periods. This is different from models that include peak time slices directly in the temporal structure and allows the modeler to choose how much detail is used to capture day to day load and iRES patterns and how much detail is used to represent critical peak periods.

The paper is structured as follows. Section 2 provides a general overview of existing adequacy methods found in literature. Next, Section 3 gives an overview of the different adequacy methods that are investigated in this paper and the methodology followed to assess their relative performance. Subsequently, Section 4 provides an in depth (mathematical) description of these adequacy methods. Section 5 presents the results and conclusions are summarized in Section 6.

2. Literature review: state-of-the-art adequacy aware energy system planning

A dynamic PRM approach is taken in the Resource planning model (RPM) [20] and Regional Energy Deployment System model (ReEDS) [21] developed by NREL. Instead of imposing a PRM constraint with fixed capacity credits throughout the planning horizon, they estimate and update the capacity credits in between consecutive milestone years. The capacity credits of iRES are estimated following the so-called 8760-based approach [22,23] which uses a separate module to determine the displacement of residual load duration curves (or net load duration curves) during 100 hours of peak residual load caused by the addition of a particular technology. A similar capacity credit estimation can be found in IEA's World Energy Model [24]. Recently, this approach has been extended to include dynamic estimations of storage technologies [25]. Using a functional form fit the authors derive an analytical expression for the capacity credit of storage that depends on the PV penetration, the storage penetration and the E/P-ratio. Their results suggest that this dynamically updating the capacity credits of storage triggers different storage investments compared to attributing a single static value. Another capacity credit estimation is introduced by Frazier [26] and uses a minimally required energy (MRE) calculation to determine the capacity credit for every E/P-ratio. In this dynamic PRM approach, more effort is spend towards accurately estimating capacity credits and the capacity credit estimation is performed for each milestone year, every iRES technology and for both the existing capacities as well as for a marginal capacity addition. Additionally, the capacity credit estimation depends on the assumptions regarding the peak period duration. For the 8760-based method the 100 peak load hours are used, but in reality this peak duration depends on the underlying power system and hence is not known a priori. Furthermore, the modeler still needs to decide on a value for the PRM. For this, often is resorted to reliability asessments of the North American Electric Reliability Corporation (NERC) [27].

As mentioned above, improving the temporal detail of the planning model is a second strategy to improve the adequacy awareness. Not only that, also the daily patterns of load and iRES generation can be captured better by including more temporal detail. As such, increasing the temporal detail is often considered a cross-cutting solution [19] that improves model accuracy on many levels. However, as pointed out by Teichgraeber et al. [28] improving the level of detail specifically during the critical peak periods increases model accuracy more than improving the overall temporal detail of the model. There are examples of power system planning models with a very detailed time slice structure, e.g., Schaber et al. [29] use 48 representative weeks with hourly resolution selected from a dataset spanning 8 years, amounting to a total of 8064 time slices. However, there is a limit as to how much temporal detail can be increased without excessively increasing the computational burden involved with solving the model. This is especially the case for models with a broad sectoral and/or geographical scope and hence, increasing the temporal detail is not always a valid option. Instead, some planning models account for special peak time slices [30,31]. This can improve the model's perception of the peak load, but a detailed estimation of technology capacity credits is likely not achieved by including only a single peak time slice. The peak time slice also depends on the underlying power system and hence, it is not likely that a single peak time slice captures the true peak for each configuration of the technology mix that can emerge.

3. Methodology

3.1. General methodology

First, we investigate the impact of the temporal detail on the adequacy awareness of planning models. To this end, we compare several limitedly detailed palnning models with a highly detailed reference model. Ideally, this reference model would contain several full-year hourly weather scenarios in a stochastic setting. However, since these scenarios would have to be included for every milestone year, this would result in a problem that is difficult to solve within a reasonable amount of time. Therefore, the reference planning model is provided by a model containing a relatively high number (i.e., 85) of representative days carefully selected out of 10 yearlong hourly profiles for load and iRES capacity factors using the optimization-based selection procedure described by Poncelet et al. [4][1] Of course, this reference model is itself an approximation and as such does not represent the 'best-possible' solution, but it should be seen as an estimate of the level of accuracy that can be obtained from a long-term planning model with a high (or at least above-average) level of temporal detail. This reference case then serves as a benchmark against which several models with a lower number of representative days are compared. In this paper, we introduce three limitedly detailed models containing respectively 5, 10 and 15 representative days. These cases are conveniently labeled as the **NoPRM cases**, since they do not include any adequacy measures.

Second, following this analysis, we investigate the impact of three different adequacy measures on planning model outcomes (Fig. 1).

[1] Please note that the same representative days are used for each milestone year, but these representative days contain information about the 10 weather year scenarios.

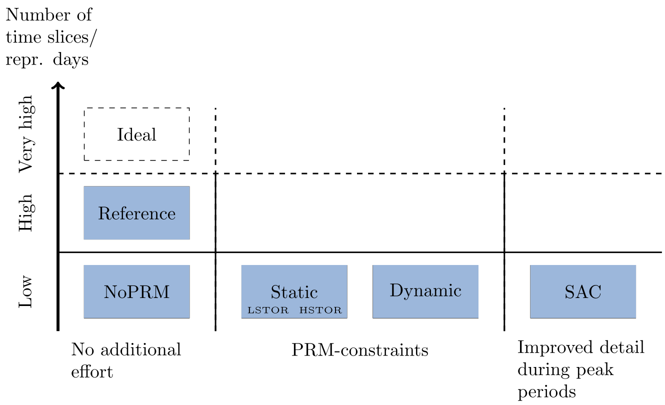

Fig. 1. Overview of the different model approaches investigated in this paper (blue rectangles). Three model categories can be distinguished, namely (i) models where no additional effort goes to increasing the adequacy awareness of the planning model, (ii) models that use PRM constraints (static or dynamic) and (iii) models that increase the level of temporal detail during peak periods. Our reference case is obtained by a planning model with 85 repr. days (which is high in the context of long-term planning models). All other cases have less temporal detail with only 5, 10 or 15 repr. days. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

1. **Static PRM -** A conventional PRM constraint with a fixed planning reserve margin is imposed on every time slice. The capacity credit of RES is determined endogenously by the capacity factor profile. The imposed capacity credit for storage is static and two situations are investigated, namely one in which the capacity credits are estimated to be low and one in which they are estimated to be high. The former is referred to as the LSTOR case, whereas the latter is referred to as the HSTOR case.
2. **Dynamic PRM -** A dynamic PRM constraint is introduced for which the imposed capacity target as well as all capacity credits are updated in between investment periods. Note that contrary to the other methods, the dynamic PRM method does not constitute a single shot optimization. The investment decisions for each milestone year are determined by sequential planning model runs which are separated by the capacity target/capacity credit updating procedure. This approach will be discussed in more detail in Section 4.3.
3. **Novel method using stochastic adequacy constraints (SAC) -** Carefully selected potential scarcity periods are included in a PRM-like constraint so that the level of detail with which the critical peak periods are modeled is increased. This approach will be discussed in more detail in Section 4.4.

The general methodology followed to compare planning models is schematically illustrated in Fig. 2. The two boxes on the left hand side reflect (i) the planning model with a high level of temporal detail, but without additional adequacy-improving measures and (ii) the limitedly-detailed planning models with (or without) additional adequacy-improving measures. In a first step, the installed capacities resulting from these models are compared with each other. Furthermore, in order to retrieve the true total system costs as well as the adequacy of the obtained installed capacities, a re-evaluation is performed in which the operational decisions of each milestone year with are evaluated with a model containing the 10 yearlong hourly weather year scenarios (the right hand side boxes).

3.2. Data and assumptions

The analysis is performed with a dynamic brownfield power-system optimization model for a system inspired by the Belgian system. The planning horizon spans the period between 2020 and 2050. The planning horizon is split up into biennial investment periods each characterized by a milestone year.

Table 1
Overview of the projected annual electricity demand, gas price and emission price for different target years.

	2020	2025	2030	2035	2040	2045
Demand [TWh]	82.2	86.3	88.7	95.6	101.8	116.9
Gas-price $[EUR/GJ]$	5.5	6.6	6.9	7.3	7.6	7.9
CO_2-price $[EUR/tonCO_2^{eq}]$	30	40	84	103	122	141

The legacy capacity, i.e., the available capacity mix at the start of the planning horizon is a simplified representation of the Belgian power system. Fig. 3 shows the assumed evolution of the legacy capacity over time. The existing nuclear power plants are assumed to follow the current plans so that a total phase-out is established by 2025. The rest of the existing capacities are assumed to gradually decrease until the year 2035.[2]

The capacity factor time series for iRES and the load time series are obtained from renewables.ninja [32,33]. The load profiles are adjusted for the temperature profiles following the methodology used in the mid-term adequacy forecast of ENTSO-E [34] and are rescaled to reach the projected annual demand in every milestone year. Changes in the load profile due to the of electrical heating and/or electric vehicles are not considered here and the demand is assumed to be inelastic.

Nine technologies are included in the long-term planning exercise, namely three conventional technologies (Nuclear, CCGT and OCGT), three intermittent technologies (Wind onshore, Wind offshore and Solar PV) and three storage technologies with different E/P-ratios (2 Wh/W, 4 Wh/W and 6 Wh/W). This stylized representation allows focusing on the fundamental effects at play. Economic and technical data for all technologies can be found in Appendix Appendix A. To incentivize investments in iRES technologies a CO_2-price is introduced and no new investments in nuclear power plants are allowed. The CO_2-price and the projected annual electric demand are based on the high renewable ambition scenario developed by EnergyVille using the Belgian TIMES model [35] and are summarized in Table 1. The assumed gas price (also presented in Table 1) is based on the new policies scenario of the International Energy Agency's World Energy Outlook 2018 [36]. Finally, a discount rate of 5% is used.

4. Model formulation: Introducing the different adequacy approaches

Section 4.1 presents the mathematical formulation of the basis planning model and the following sections explain how the three different adequacy-improving approaches relate to this basis model. Please note that although some approaches require more effort of the modeler than others, all three approaches presented here are generally valid and can be applied in any type of power-system optimization model because the backbone of these models are very similar in terms of the structure of the objective function and the imposed constraints.

4.1. Basis planning model

The basis model takes the form of a normal power-system optimization model and is based on the LUSYM invest model [37]. The model

[2] It must be noted that this is not necessarily the case in reality, and that for real scenario analyses that aim to provide insights in the energy transition for Belgium, the retirement of existing capacity needs to be represented as accurately as possible. However, since our aim is to investigate the impact of different adequacy methods in a rather methodological analysis, in this work the assumption is made that within a timespan of 15 years all existing capacity is retired.

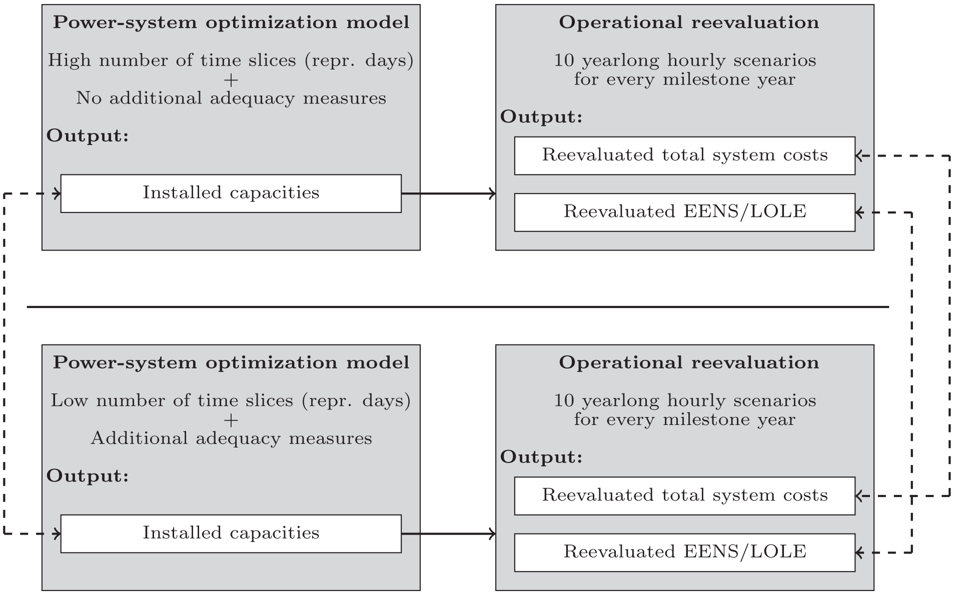

Fig. 2. Schematic overview of the methodology used to compare planning models with high number of time slices and no additional adequacy methods and planning models with a low number of time slices and additional adequacy methods.

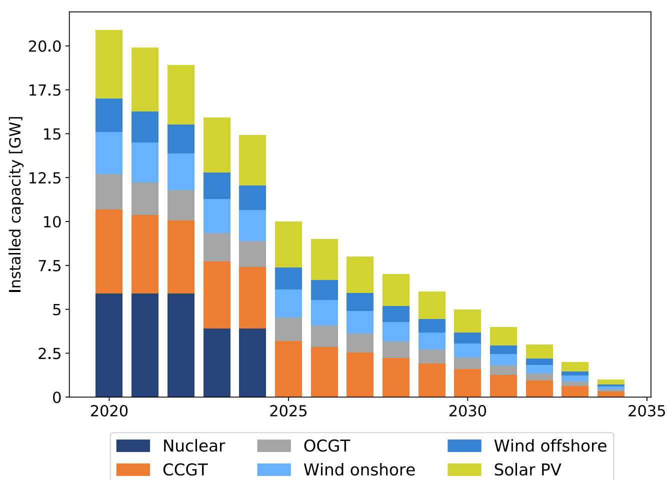

Fig. 3. The available legacy capacity at the start of the planning model. Values are based on a simplified representation of the Belgian power system and all legacy capacity is assumed to have retired in 2035. The existing capacity mix contains 6 technologies, namely nuclear power plants, combined-cycle gas turbines (CCGTs), open-cycle gas turbines (OCGTs), onshore wind turbines, offshore wind turbines and solar PV installations.

minimizes the total cost of maintaining and operating a power system with a projected electricity demand. The full model description is included in Appendix Appendix B. Here, we restrict the description to an overview of the costs included in the objective function and the power balance constraint.

The objective function expresses a minimization of the discounted total system costs, which is expressed in Eq. (1) as the sum over all milestone years y of investment costs (c_y^{inv}), fixed operation and maintenance costs (c_y^{fom}), variable operation and maintenance costs (c_y^{vom}), generation costs (c_y^{gen}), and load shedding costs (c_y^{ens}). The factor DF_y represents the discount factor associated with milestone year y.

$$Min \sum_{y \in Y} DF_y \cdot \left(c_y^{inv} + c_y^{fom} + c_y^{vom} + c_y^{gen} + c_y^{ens} \right) \tag{1}$$

Eq. (2) expresses the balance between supply and demand for every milestone year y, representative day i and time step t.

$$\sum_{g \in G} \sum_{v \in V} gen_{g,v,y,i,t} + \sum_{s \in S} \sum_{v \in V} dc_{s,v,y,i,t} + ens_{y,i,t} = D_{y,i,t} + \sum_{s \in S} \sum_{v \in V} ch_{s,v,y,i,t} \qquad \forall y \in Y; i \in I; t \in T \tag{2}$$

Other constraints include the activity constraints that limit the activity (generation, charge or discharge) of a technology to the installed capacity, the constraints tracking the state-of-charge of the storage technologies, and the constraints expressing the storage limits. For the sake of brevity, we excluded the mathematical formulation of these constraints in the main body of the paper. However, the full mathematical description can be found in Appendix Appendix B.

4.2. Static planning reserve margin constraints

The first approach uses a static PRM constraint to improve the adequacy awareness of the basis planning model. This constraint can be written as follows:

$$\sum_{g \in G_D} \sum_{v \in V} cap^{av}_{g,v,y} + \sum_{g \in G_R} \sum_{v \in V} CF_{g,i,t} \cdot cap^{av}_{g,v,y} + \sum_{s \in S} \sum_{v \in V} CC_s \cdot cap^{av,P}_{s,v,y} \geq (1 + PRM) \cdot D_{y,i,t} \qquad \forall y \in Y; i \in I; t \in T \tag{3}$$

A fixed margin of 10% is used for all milestone years, which is within the range of commonly encountered PRM values [27]. The contributions of iRES technologies are determined by the capacity factor time series and the capacity credit of storage is exogenously imposed. Note that although the static PRM constraint is imposed on every time step included in the representative days, only the PRM constraint at the residual peak time step is binding. Hence, only a single step determines the imposed capacity target and the imposed capacity credit for iRES technologies.

Furthermore, two sets of storage capacity credits are imposed, namely one in which the capacity credits of the storage technologies are relatively high (HSTOR) and one in which the capacity credits of the storage technologies are relatively low (LSTOR). For the HSTOR-case the capacity credits of the storage technologies with an E/P-ratio of 2, 4 and 6 are respectively 0.3, 0.4 and 0.5. For the LSTOR case these values are 0.1, 0.2 and 0.3.

4.3. Dynamic planning reserve margin constraints

The dynamic PRM approach is the only approach that is not solved with a single-shot optimization, i.e., one single optimization that determines all investment decisions at once. Instead, the investment decisions of each milestone year are sequentially obtained by consecutively solving the basis planning model with updated information regarding the required capacity target and the capacity credit for the different technologies.

In each of the basis model runs a PRM-like constraint is imposed that requires the model to invest in sufficient firm capacity in year y so that an adequate capacity mix in year $y+1$ is obtained. For example, to determine the investment decisions for the first milestone year (2017), a capacity constraint is imposed so that the investments made in 2017 result in an adequate capacity mix in the second milestone year (2022). To determine the investment decisions in the second milestone year (2022), a capacity constraint is imposed so that the investments made in 2022 result in an adequate capacity mix in the third milestone year (2024), etc. Mathematically the dynamic PRM constraint can be written as:

$$\sum_{g\in G_D} cap^{av}_{g,y,y+1} + \sum_{g\in G_R} CC_g \cdot cap^{av}_{g,y,y+1} + \sum_{s\in S} CC_s \cdot cap^{av,P}_{s,y,y+1} \geq CapTarget \tag{4}$$

where $CapTarget$ is the estimated additional firm capacity required to be installed in year y so that the capacity mix is adequate in year $y+1$. To estimate this capacity target as well as the marginal capacity credits of the different technologies, a separate external module is used. This external module uses a highly detailed operational model to determine how much additional firm capacity is required to be installed in year y so that an adequate system is obtained in year $y+1$. Given the already existing capacities, this module is also able to estimate the capacity credits of iRES and storage technologies.

Of course, this approach requires almost as much evaluations of the basis planning model and the separate module as there are milestone years (Y-1 evaluations for Y milestone years). This inevitably increases the computational time required to obtain the final solution, but does not really increase the complexity of the basis planning model.

4.4. Novel model formulation using stochastic adequacy constraints

4.4.1. Principle behind the SAC approach

The rationale behind the SAC approach is that we increase the level of detail with which critical peak periods are modeled, so that the planning model can endogenously determine the optimal amount of firm capacity to be installed. Other approaches have tried to pursue this strategy through adding more representative days or by adding a special peak day. One challenge with this approach is that out of all time slices across all weather year scenarios, the moments at which capacity becomes scarce are usually not confined to a single day. A second challenge is that these scarcity moments depend on the underlying capacity mix.

In this regard, the approach proposed here takes a different pathway to increase the level of detail in the peak periods. We include a new constraint in which the power balance is evaluated for a carefully selected set critical peak periods. As such, the constraint that evaluates the power balance for the representative days (which determine the costs associated with generating electricity) is different from the power balance constraint used to make the trade-off between installing additional capacity and load shedding. By explicitly decoupling these aspects, the modeler is free to choose the level of detail with which both aspects are modeled. This is beneficial as it provides flexibility as to how computational efforts are allocated within the model design. Furthermore, these critical peak periods do not have to fit a uniform structure, i.e., the length of each period can be different for each period.

The challenge for this approach is that the critical peak periods need to be selected before actually solving the planning model. It is the responsibility of the modeler to make an educated choice of which periods to include. This choice could be supported by (simplified) model runs for various compositions of the capacity mix to identify likely scarcity periods. For the purposes of our analysis, we use the capacity mix emerging from the NoPRM case with 5 repr. days to investigate the relevant scarcity periods with a detailed operational model.

4.4.2. Mathematical formulation

To implement the SAC-approach, the basis model needs to be restructured. The most important features of the new planning model formulation are summarized here. First and foremost, the traditional power balance constraint is changed so that load shedding does not contribute to this constraint. Instead, a dummy generation term is added, which acts as a slack variable for this constraint. This slack variable is necessary to make sure that the dispatch situations reflected by the representative days do not determine the required firm capacity of the system. Without this slack variable it would be possible for this (low detailed) power balance constraint to drive investments in firm capacity through the activity constraints, i.e., the constraints that limit the activity (generation, charge or discharge) of a technology to the installed capacity, which is exactly what we want to avoid.

$$\sum_{g\in G_D}\sum_{v\in V} gen_{g,v,y,i,t} + \sum_{g\in G_R}\sum_{v\in V} gen_{g,v,y,i,t} + \sum_{s\in S}\sum_{v\in V} dc_{s,v,y,i,t} = D_{y,i,t} + \sum_{s\in S}\sum_{v\in V} ch_{s,v,y,i,t} - gen^{dummy}_{y,i,t} \quad \forall y\in Y; i\in I; t\in T \tag{5}$$

The same activity constraints as in the basis model are included as well as the constraints tracking the state-of-charge of the storage technologies, and the constraints expressing the storage limits.

Second, a new constraint is added that evaluates the power balance for ever time slice h in each carefully selected peak period p. This constraint resembles the traditional power balance constraint and includes the possibility of load shedding. It is referred to as the stochastic adequacy constraint and takes the following form:

$$\sum_{g\in G_D}\sum_{v\in V} cap^{av}_{g,v,y} + \sum_{g\in G_R}\sum_{v\in V} gen^{SAC}_{g,v,y,p,h} + \sum_{s\in S}\sum_{v\in V} dc^{SAC}_{s,v,y,p,h} = D^{SAC}_{y,p,h} + \sum_{s\in S}\sum_{v\in V} ch^{SAC}_{s,v,y,p,h} - ens^{SAC}_{y,p,h} \quad \forall y\in Y; p\in P; h\in H(p) \tag{6}$$

The updated expression for the objective function is reflected in (7) and differs in two aspects from the objective function in the basis model. First, a dummy generation cost slightly higher than the marginal cost of the peak generator is added to make sure that the $gen^{dummy}_{y,i,t}$-variable is only used when all other means to satisfy the demand (reflected in the representative days) are exploited. Second, the cost associated with load shedding is evaluate for the time slices h in the critical peak periods P.

$$Min \sum_{y\in Y} DF_y \cdot \left(\underbrace{c^{inv}_y + c^{fom}_y + c^{vom}_y + c^{gen}_y + c^{dummy}_y}_{\text{Evaluated for all representative days} I} + \underbrace{c^{ens}_y}_{\text{Evaluated for all peak periods} P} \right) \tag{7}$$

Where,

$$c^{dummy}_y = N^{PY}_y \cdot \sum_{i\in I} W_i \cdot \left[\sum_{t\in T} MC_{\hat{g},y} \cdot gen^{dummy}_{y,i,t} \cdot \Delta \right] \quad \forall y\in Y \tag{8}$$

and

$$c^{ens}_y = N^{PY}_y \cdot \sum_{p\in P} W^{SAC} \cdot \left[\sum_{h\in H} \left(VOLL - MC_{\hat{g},y} \right) \cdot ens^{SAC}_{y,p,h} \right] \tag{9}$$

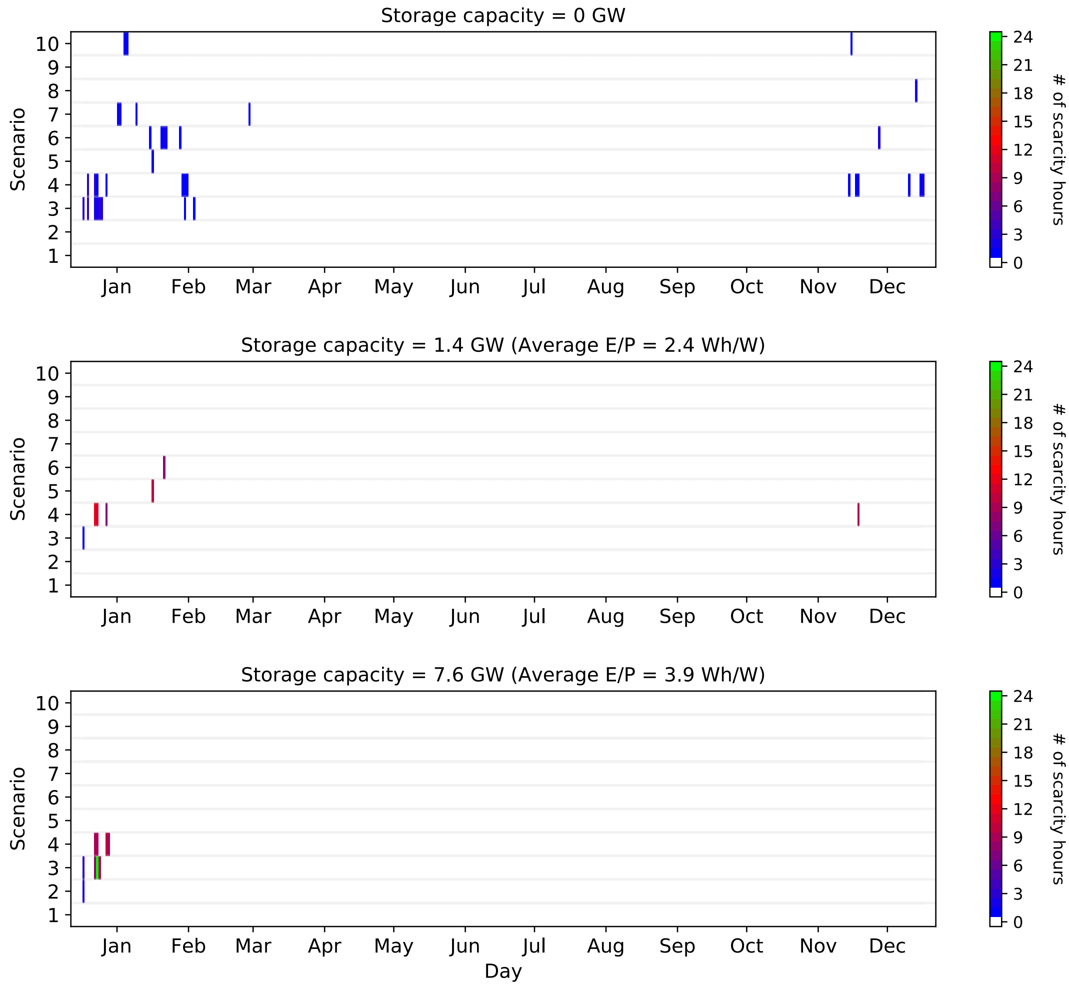

Fig. 4. Distribution of the scarcity periods across weather year scenarios for different storage penetrations. The rectangles indicate the days in which scarcity emerges and the colors reflect the duration of the scarcity period for that day.

Please note that the cost associated with load shedding is equal to the difference between the value of lost load and the marginal cost of the peak generator in a particular year. This is to avoid double counting of the cost associated with generating electricity. Recall that $\sum_{i\in I} W_i = 365$ and that there is an operational cost associated every hour of every year represented in the representative days. As such, the cost associated with the ens-variable should only reflect the capacity component included in the VOLL. Furthermore, the weight W^{SAC} assigned to each critical peak moment reflects the duration of the peak moment as well as the probability of it occurring. Since in this analysis we select peak periods with hourly resolution from 10 equiprobable weather year scenarios these weights are equal to $\frac{1h}{10} = 0.1h$. N_y^{PY} reflects the number of years within an investment period adjusted for the discounting to the milestone year of that investment period. Since each investment period consists of two consecutive years of which the first is the milestone year, and given a discount rate of 5% the discounted number of years in one investment period is equal to approximately 1.95 years.

Finally, the traditional generation limits need to be respected for renewable technologies (restricting the variable $gen^{SAC}_{g,v,y,p,h}$) and the state-of-charge of each storage unit needs to be tracked across each peak period p.

$$e^{SAC}_{s,v,y,p,h+1} = e^{SAC}_{s,v,y,p,h} + \sqrt{\eta_{s,v}} \cdot ch^{SAC}_{s,v,y,p,h} - \frac{dc^{SAC}_{s,v,y,p,h}}{\sqrt{\eta_{s,v}}}$$
$$\forall s \in S; v \in V; y \in Y; p \in P; h \in H(p) \tag{10}$$

One important assumption needs to be made regarding the state-of-charge of the storage units at the beginning of each peak period. Given that the stochastic adequacy constraint only determines the usage of the ens-variable, we assume that at the start of each period the storage units are filled to their maximum energy capacity. This assumption is reflected in Eq. (11).

$$e^{SAC}_{s,v,y,p,1} = cap^{E,av}_{s,v,y} \qquad \forall s \in S; v \in V; y \in Y; p \in P \tag{11}$$

4.4.3. Selecting critical peak periods

Using the classification proposed by Teichgraeber et. al [28], the selection procedure used in this paper can be described as a simple selection process, with an optimization-based identification measure, i.e., the dual variable associated with the power balance constraint of the highly detailed operational model including 10 weather year scenarios. Whenever this dual variable(which reflects the electricity price) exceeds the marginal cost of the peak generator at a time step, that time step is considered part of the critical peak period. To get a sense of which capacity mixes are useful to investigate in this regard, a simplified planning exercise (with only a handful representative days) could be used to identify likely ranges of iRES and storage penetrations.

This paper re-evaluates the capacity mixes emerging from the No-PRM case with 5 repr. days with the highly detailed operational model to identify critical peak periods. Fig. 4 shows the distribution of the scarcity hours across weather years and days of the year for three different capacity mixes. These capacity mixes are mostly characterized by different storage penetrations. As such, Fig. 4 shows the impact of stor-

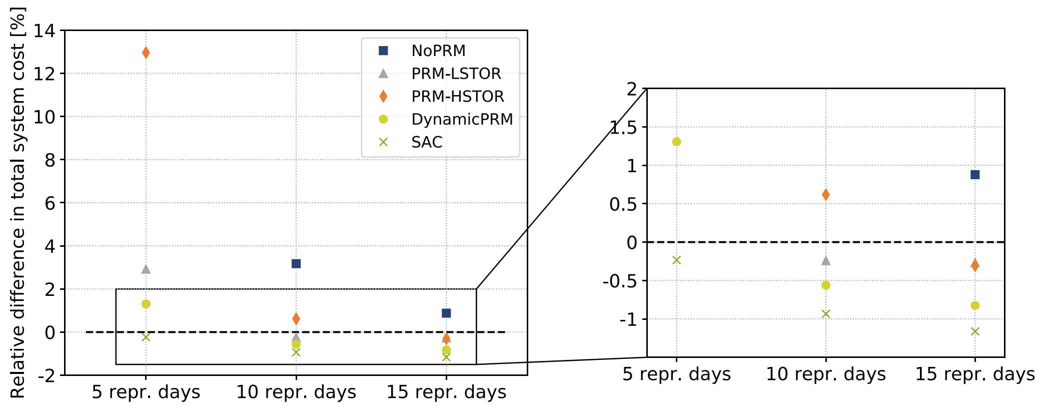

Fig. 5. Difference in re-evaluated costs associated with the different adequacy measures relative to the re-evaluated cost of the reference case (dashed black line).

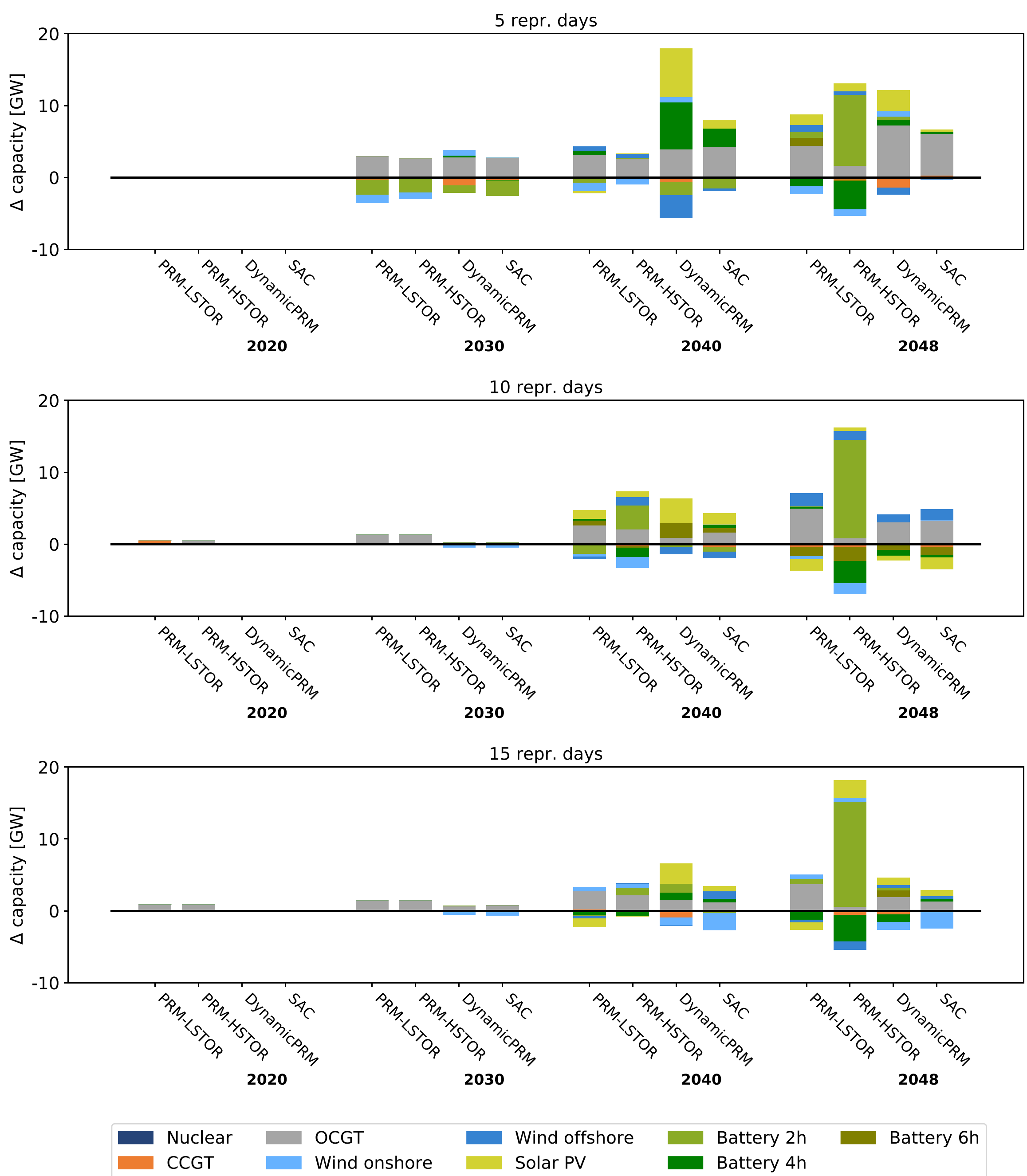

Fig. 6. The differences in capacity mix with respect to the time specific NoPRM cases for several milestone years.

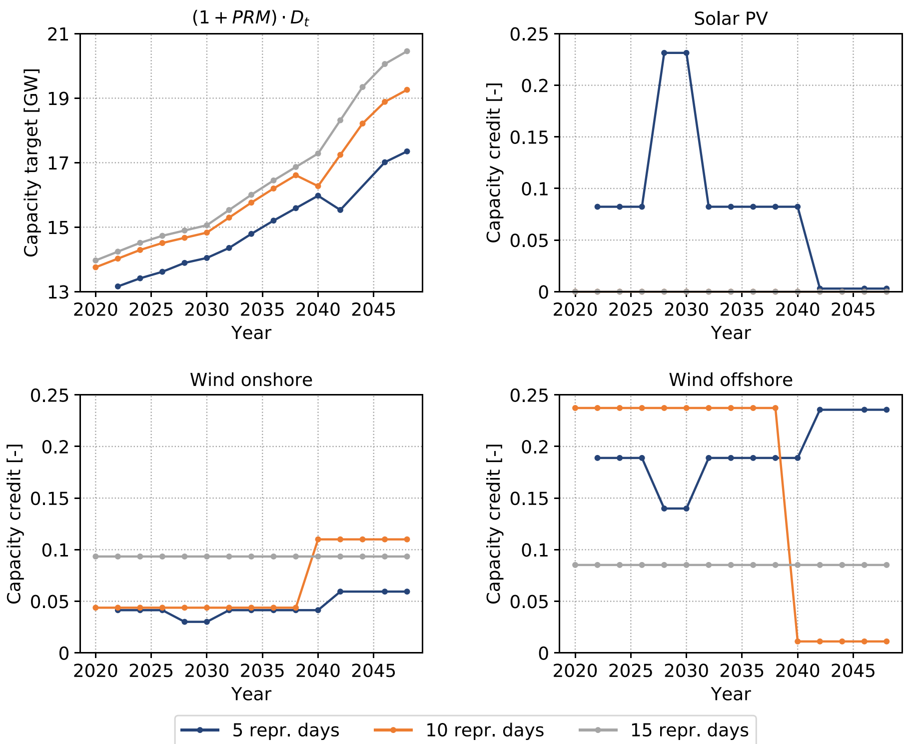

Fig. 7. Capacity targets and iRES capacity credits across milestone years for each set of representative days as determined by the static PRM constraint.

age on the distribution of scarcity hours. As can be seen, without any storage (top panel) the scarcity moments are spread out across the entire winter period across various weather years, while as more storage is added these peak periods are more concentrated in time and in a couple of weather year scenarios.

5. Results and discussion

First, Section 5.1 presents an overview of the different model approaches and their outcomes, and discusses the impact of the temporal detail on the adequacy awareness of planning models. Next, Section 5.2 takes a closer look at the advantages and disadvantages associated with each method. Finally, Section 5.3 contains a discussion regarding model complexity.

5.1. Relevance of taking additional measures to increase the adequacy awareness of planning models

5.1.1. Re-evaluated total system costs

Fig. 5 shows the differences in re-evaluated costs associated with the different cases relative to the re-evaluated cost of the reference case. As can be seen, regardless of the number of representative days, taking additional adequacy measures decreases overall system costs (especially when the temporal detail is very low). The SAC approach here seems to consistently perform better than other approaches, with a cost reduction of 95%[3] for the case with 5 repr. days, 4% for the case with 10 repr. days and 2% for the case with 15 repr. days. The dynamic PRM-approach also performs well, although the performance depends more on the included temporal detail (the re-evaluated costs increase more strongly as the number of representative days decreases).

Additionally, Fig. 5 shows that increasing the number of representative days to 85 does not necessarily perform better (in terms of system costs) than planning models with lower temporal detail but with additional adequacy measures. This suggests that it could be more beneficial to increase the level of detail specifically in the critical peak periods rather than in the overall temporal structure of a planning model. Note that the relative cost differences between the models mostly depends on the assumption regarding the VOLL, i.e., the re-evaluated costs are driven by excessive load-shedding rather than suboptimal operation of technology mixes.

5.1.2. Adequacy awareness

Table 2 shows the re-evaluated reliability metrics associated with the obtained capacity mixes. To guide the reader through the results the reliability metrics are given different color codes based on the LOLE values. All situations for which the LOLE is greater than 10h are depicted in red and all situations for which the LOLE is smaller than 1h are depicted in blue. LOLE values of 10h and 1h are no strict boundaries, but can aid in identifying capacity mixes with excess load shedding and those with a shortage of load shedding (scarcity hours needed to recover the fixed costs of the peak generators).

From Table 2 we can see that without adequacy measures (i.e, the NoPRM cases), the obtained capacity mixes are inadequate across the entire planning horizon and the inadequacies tend to be higher as less temporal detail is included. These inadequate capacity mixes are mainly caused by the underestimation of the system's peak load due to the simplified temporal representation. However, even for the reference case there is still substantial load shedding present. The implication of this is that when the temporal detail of a planning model is limited, additional adequacy measures are necessary to increase the adequacy of the obtained capacity mix.

Regarding the static PRM constraints, it can be observed that the adequacy metrics are also dependent on the temporal detail. Indeed, the PRM value of 10% is insufficient to obtain an adequate capacity mix for the case with 5 repr. days. However, for the cases with 10 and 15 repr. days, almost all adequacy metrics are reduced to zero, which suggests that the PRM of 10% results in overcapacity. Furthermore, for the HSTOR the capacity mixes become more inadequate towards the

[3] This point is not shown on Fig. 5 since this would decrease the readability of the graph.

Table 2
Re-evaluated reliability metrics associated with the different cases for several milestone years. The situations with a LOLE lower than 1h are depicted in blue and the situations with a LOLE greater than 10 h are depicted in red.

			2024	2028	2032	2036	2040	2044	2048
	85 repr. days	EENS [GWh]	2.91	5.28	5.67	10.49	19.31	69.88	70.61
		LOLE [h]	10.8	18.9	15.6	21.2	26.4	48.3	48.4
NoPRM	5 repr. days	EENS [GWh]	75.57	43.87	406.00	1792.66	2043.79	1786.41	3965.86
		LOLE [h]	170.9	86.4	356.0	1125.4	1120.7	559.4	1050.9
	10 repr. days	EENS [GWh]	4.09	2.74	12.74	74.26	68.50	57.40	406.52
		LOLE [h]	17.6	11.2	24.7	92.2	69	31.4	157.3
	15 repr. days	EENS [GWh]	0.76	8.61	36.01	36.62	38.90	60.71	59.70
		LOLE [h]	4.7	25.7	61.7	56.2	49.7	48.6	47.9
PRM-LSTOR	5 repr. days	EENS [GWh]	1.12	0.51	17.20	59.74	77.34	85.03	112.64
		LOLE [h]	6.3	2.7	36.5	78.0	77.8	47.5	57.2
	10 repr. days	EENS [GWh]	0	0	0	0.20	0.04	0.35	0.31
		LOLE [h]	0	0	0	0.7	0.1	0.3	0.3
	15 repr. days	EENS [GWh]	0	0	0.07	0.13	0	0	0
		LOLE [h]	0	0	0.4	0.5	0	0	0
PRM-HSTOR	5 repr. days	EENS [GWh]	1.12	0.51	17.51	105.96	171.31	264.09	1756.48
		LOLE [h]	6.3	2.7	36.7	102.8	141.3	131.3	536.5
	10 repr. days	EENS [GWh]	0	0	0	0.42	0.52	3.03	233.29
		LOLE [h]	0	0	0	0.8	0.9	3.4	100.6
	15 repr. days	EENS [GWh]	0	0	0.08	0.15	0.03	2.33	17.41
		LOLE [h]	0	0	0.4	0.6	0.1	2.0	14.3
Dynamic PRM	5 repr. days	EENS [GWh]	0.98	0.80	30.10	1.11	3.03	1.30	2.36
		LOLE [h]	5.8	3.5	47.5	1.3	3.7	1.4	2.9
	10 repr. days	EENS [GWh]	0.98	2.74	9.34	1.47	1.23	5.11	1.61
		LOLE [h]	5.8	11.2	20.7	2.6	1.2	4.9	1.5
	15 repr. days	EENS [GWh]	0.76	4.44	17.37	2.82	1.14	1.32	1.47
		LOLE [h]	4.7	17.0	35.0	4.8	2.3	1.6	1.5
SAC	5 repr. days	EENS [GWh]	1.47	1.22	1.42	1.76	1.09	1.54	1.69
		LOLE [h]	8.0	5.7	5.3	4.3	1.3	1.8	1.7
	10 repr. days	EENS [GWh]	1.47	1.11	1.75	1.51	1.11	1.51	2.30
		LOLE [h]	8.0	5.6	5.8	4.5	1.7	1.4	2.6
	15 repr. days	EENS [GWh]	1.47	1.24	1.39	1.65	1.49	1.55	3.55
		LOLE [h]	8.0	5.7	5.5	4.8	3.6	1.6	3.0

end of the planning horizon due to the overestimation of the capacity credits for storage. These results illustrate that the performance of static PRM constraints is highly dependent on temporal detail and the chosen parameters.

Both the dynamic PRM approach as well as the SAC approach effectively improve the adequacy awareness of the planning model regardless of the temporal detail, although the dynamic PRM approach suffers from occasional inadequacies for individual milestone years. The SAC approach is the most consistent throughout the planning horizon.

5.1.3. Technology choices

Fig. 6 displays the capacity mix differences caused by the different adequacy methods compared to each time specific NoPRM case. As can be seen, most adequacy methods result in more OCGT investments which indicates that the methods effectively increase the firm capacity in the system. Regarding the PRM-LSTOR case with 5 repr. days, the amount of additional OCGTs installed is lower than for the SAC approach, while with 10 and 15 representative days the opposite is true. This confirms our hypothesis that the imposed PRM of 10% does not result in an adequate capacity mix with 5 representative days, while it causes overinvestments for the 10 and 15 representative days. Regarding the PRM-HSTOR cases, Fig. 6 shows lower investments in OCGTs and the larger investments in the 2 Wh/W storage technology towards the end of the planning horizon. This technology bias is a result of the higher capacity credits for storage technologies and the cause of the inadequacies shown in Table 2. Finally, it can be observed that the dynamic PRM method also occasionally shows technology biases. This is most visible in the model runs with 5 representative days, where larger changes in the iRES and storage mix emerge.

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5.2. A closer look at the different adequacy measures

5.2.1. Static PRM constraints

Fig. 7 shows the imposed capacity targets over the planning horizon for every set of representative days as well as the endogenously determined capacity credits for the renewable technologies. As can be seen, both the capacity target and the capacity credits are strongly dependent on the temporal detail. By increasing the number of representative days, the imposed capacity targets (for a constant 10% PRM) are higher due to the improved estimation of the peak load. Furthermore, for the cases with 5 and 10 repr. days the capacity target experiences a sudden drop in 2042 and 2040 respectively. This indicates that the time slice at which the PRM constraint becomes binding changes due to increased iRES investments.

Regarding the capacity credits for iRES technologies, the impact of the temporal representation is also clearly visible. For each milestone year only a single time slice determines the capacity credit and as such, depending on which time slice is binding, the value of the capacity credit can vary. It is unlikely that a single time slice is representative for the entire set of scarcity moments that can occur. This dependency of capacity credits and capacity targets on one particular times slice diminish the transparency of the static PRM approach.

Regarding the storage technologies, Fig. 6 showed that for the HSTOR case large technology biases are present. Indeed, a bias of around 17 GW in favor of the 2 Wh/W-technology is observed in the HSTOR case. To elaborate on this effect, it is important to understand that a PRM constraint seeks to reach the imposed capacity target in the cheapest

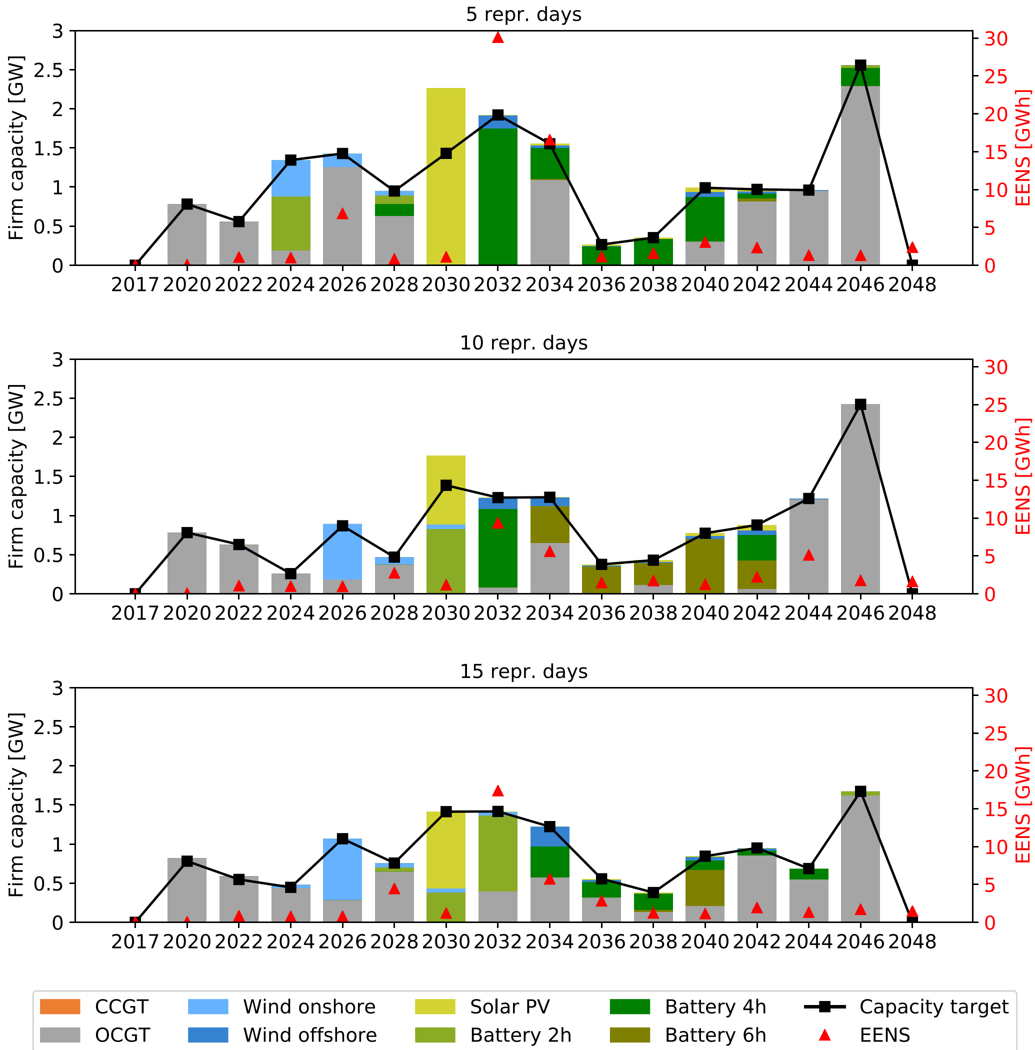

Fig. 8. The firm capacity additions (left axis) for each milestone year due to the imposed capacity target (black line) for the dynamic PRM method. The red triangles represent the re-evaluated EENS (right axis) for the capacity mix at each milestone year. We observe increased levels of EENS for each milestone year that succeeds a year in which there are substantial investments in iRES or storage technologies (see e.g., 2030 and 2032). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

way possible. The technology used to reach this target is then the technology for which the unit fixed cost, compensated for the capacity credit, and the change in operational costs by adding a unit of capacity is the lowest, or mathematically, the technology g for which $\frac{FC_g}{CC_g} + \Delta OPEX$[4] is minimal. The 2 Wh/W-technology has the lowest fixed cost per unit of power capacity and has a capacity credit of 0.3 in the HSTOR case, which is enough to prioritize the 2Wh/W technology over all other storage technologies and the peak technology.[5] This effect is of course not desirable and should be avoided when imposing capacity credits exogenously.

5.2.2. Dynamic PRM constraints

Fig. 8 shows the imposed capacity target and the firm capacity used to reach this target for each milestone year for the three sets of representative days. Additionally, the red triangles show the re-evaluated EENS associated with the resulting capacity mix in each milestone year. As can be seen, the imposed capacity target is met by a combination of OCGTs, iRES and storage technologies and the re-evaluated EENS values are relatively low for most milestone years. However, the EENS values tend to be larger for the years that are preceded by iRES or storage investments. For example, in 2030 the added firm capacity is mostly characterized by solar PV and the 2 Wh/W storage technology, after which the EENS spikes in 2032 for all three sets of representative days. This suggests that the firm capacity additions of 2030 in fact are insufficient to reach an adequate capacity mix in 2032.

The reason for this can be found in Fig. 9, which displays the capacity credits for iRES and storage technologies as determined by the external module for the case with 5 repr. days. As a result of the retiring capacity legacy, the solar capacity decreases which causes the estimated capacity credit of solar PV to increase towards 2030. Although, these capacity credit calculations are accurate, they are only valid for a marginal addition of solar PV capacity. The capacity credit of iRES decreases strongly when more capacity is installed and as such, when large amounts of solar PV are installed in a single milestone year, the imposed capacity credit is actually an overestimation of the true capacity credit. As a result, the model overestimates the amount of firm capacity that is installed, which

[4] Note that for storage, renewable and conventional technologies that are not the peak technology the change in operational costs by adding (and optimally operating) a unit of capacity is negative (or equal to zero). For the peak technology the change in operational costs by adding a unit of capacity is zero as the capacity is added at the end of the merit-order.

[5] Note that this effect can also occur for renewable technologies if capacity credits are imposed exogenously.

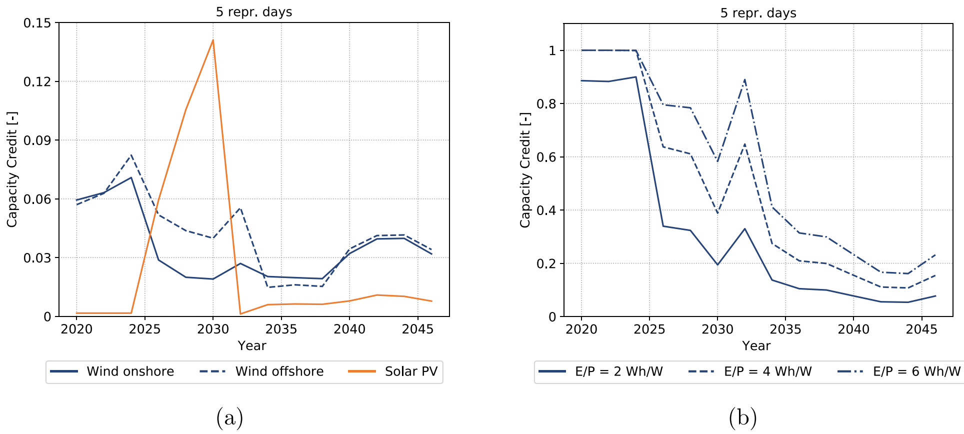

Fig. 9. Capacity credits as determined by the capacity credit estimation of the dynamic PRM method for iRES technologies (a) and storage technologies (b).

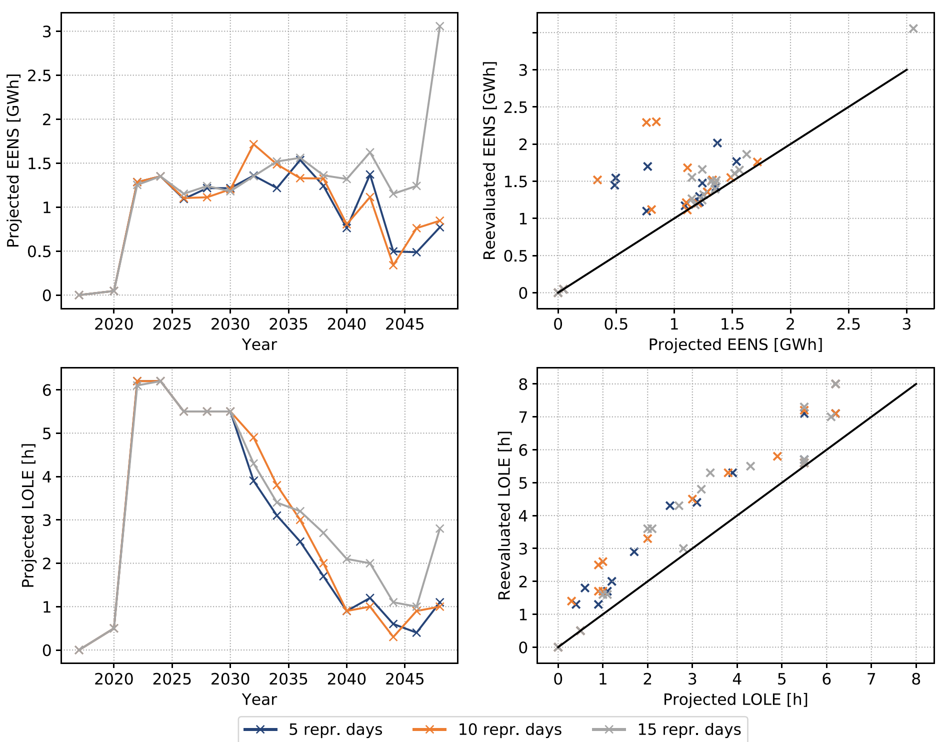

Fig. 10. The left panels show the projected reliability metrics for the three sets of representative days as determined by the stochastic adequacy constraints. The right panels display the relationship between the projected and the reevaluated reliability metrics for all milestone years.

results in inadequacies. Note that this overestimation is adjusted during the next capacity credit update in 2032, causing a strong decrease in the capacity credit. A similar effect can be seen for the storage technologies in Fig..

5.2.3. *Stochastic adequacy constraints*

The results presented in Section 5.1 suggest that the stochastic adequacy constraints, i.e., increasing the level of detail during critical peak periods, performs the best in approaching a cost-optimal solution. Also regarding generation adequacy the SAC approach is the most consistent, with LOLE values that never exceed 8 hours. An additional benefit of this approach is that an estimate of the generation adequacy can be obtained without having to re-evaluate the obtained solution with a detailed operational model.[6] Fig. 10 shows the projected EENS and LOLE values for each milestone year as well as the relationship between these projected values and the re-evaluated values. As can be seen, the projected metrics underestimate the re-evaluated metrics which indicates that the critical peak periods used do not capture all scarcity events associated with the obtained capacity mix. Nevertheless, the approximation seems to be reasonably accurate for several milestone years. The main challenge for the SAC approach is ensuring a sufficiently detailed/accurate representation of the potential scarcity periods without perfect knowledge of the resulting capacity mix.

5.3. *Model complexity*

Fig. 11 a displays the relative total system cost and the computational time associated with each model run. What stands out is that for the dynamic PRM approach the computational time is much higher than for the other cases, i.e., between 6700 s and 6900 s. The reason for this is that the dynamic PRM approach relies on multiple planning model runs, rather than one single-shot optimization. Furthermore, the majority of the computational time is spend solving the external module for estimating the capacity targets and the capacity credits, which consists of a highly detailed operational model. Overall computational time can be decreased by speeding up this operational model by more

[6] Note that all other model approaches have a too low temporal resolution to provide accurate estimates for the reliability metrics.

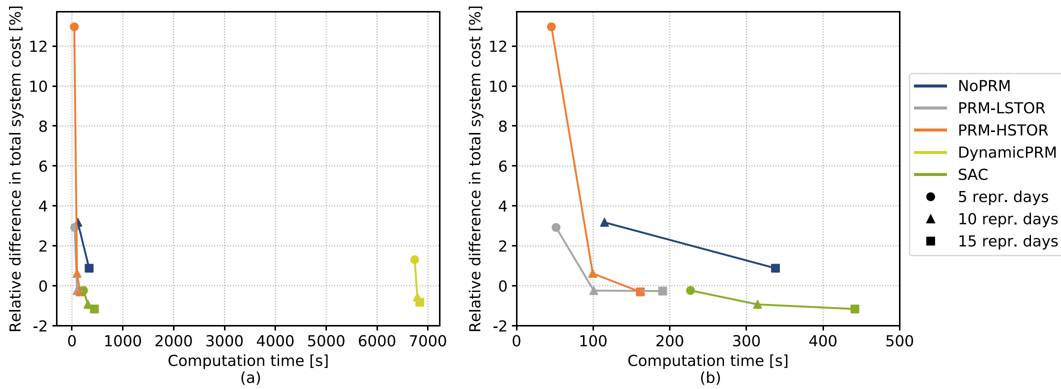

Fig. 11. The obtained re-evaluated system costs against the computational time associated with solving each planning model design.

efficiently evaluating this operational model, e.g., by reducing the number of weather years used or by including only the months that likely contain the peak periods in the analysis (e.g., the winter months).

Fig. 11 b shows that the static PRM model runs are the fastest, but their accuracy is volatile and depends on the level of temporal detail, the imposed PRM and the assumed storage capacity credits. Furthermore, the capacity mixes with the lowest system costs are obtained by the SAC-approach due to the increased level of detail with which the peak periods are modeled. This of course requires more computational effort compared to the other approaches resulting in higher computational times. A more efficient selection of the critical peak periods could potentially decrease the computational burden while maintaining roughly the same accuracy, but the main challenge remains the limited prior knowledge about the resulting capacity mix.

6. Conclusion

This paper investigated three different approaches to deal with inadequacies resulting from a simplified temporal representation, namely (i) a static PRM constraint, (ii) a dynamic PRM constraint and (iii) a novel approach that increases the level of detail with which critical peak periods are modeled (SAC approach). Our results show that, from the three approaches investigated in this paper, the novel SAC approach consistently results in more cost-optimal technology choices as the resulting re-evaluated system costs are up to 1% smaller than the reference model. This highlights the importance of peak periods for the endogenous valuation of different technologies. As such, if the temporal structure already approximates load and iRES duration curves relatively well (which is the case for, e.g., a small set of representative days), increasing the level of detail in likely peak periods should be prioritized. Another benefit of this approach is that the re-evaluated adequacy metrics are in line with the projected adequacy metrics by the model. The main challenge is identifying potential critical peak periods. This requires some additional effort from the modeler prior to running the planning model, but this can be achieved by using results of a simplified planning model in combination with a detailed operational model (as done in this paper). Therefore, the SAC approach remains the preferred approach to increase the adequacy awareness of a planning model.

The dynamic PRM approach also results in mostly adequate capacity mixes, although for some milestone years high levels of EENS are observed. This is caused by overestimations of the firm capacity provided by certain technologies. These inaccuracies do not emerge through flawed capacity credit estimations as such, but rather the lack of an iterative procedure to align the capacity credit with the technology penetration within a milestone year. This effect is always present when capacity credits are exogenously imposed for technologies with capacity credits that depend on the underlying power system. Therefore, care should be taken with these exogenously imposed capacity credits, especially when large investments in storage or iRES technologies occur in a single milestone year. Furthermore, since the dynamic PRM approach requires multiple sequential planning model runs, which inevitably increases computational time.

Static PRM constraints have the benefit of simplicity. However, an important pitfall of this approach is that it is not as transparent as the concept might suggest. The performance is highly dependent on the level of temporal detail of the model and the extent to which peak demand in underestimated. As such, the choice of the PRM should be tailored to the adopted temporal structure. Furthermore, capacity credits of iRES technologies are often determined endogenously by the capacity factor at the binding time slice. This dependency on a single time slice might not result in accurate capacity credit estimates when the level of temporal detail is low, which further diminished the transparency of the static PRM constraints. Finally, static exogenously imposed capacity credits can lead to strong technology biases (in our case study a technology bias of 17 GW for the 2 Wh/W storage technology was observed).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Tim Mertens: Conceptualization, Methodology, Software, Writing – original draft, Formal analysis, Visualization. **Kenneth Bruninx:** Conceptualization, Methodology, Writing – review & editing. **Jan Duerinck:** Supervision, Writing – review & editing. **Erik Delarue:** Supervision, Writing – review & editing.

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Appendix A Technology-specific data

Tables A1, A2 and A3

Table A1
Technology-specific parameters of the conidered conventional, renewable and storage technologies used in this paper. Data is roughly based on [38] and [39].

Technology	Lead time [years]	Lifetime [years]	Efficiency [%]
Nuclear	8	60	33
CCGT	4	25	51
OCGT	2	25	34
Wind onshore	2	25	-
Wind offshore	2	25	-
Solar PV	2	25	-
Battery	1	15	85 (round-trip)

Table A2
Technology-specific costs of the considered conventional and renewable technologies used in this paper. Data is roughly based on [38].

Technology	**Investment [ke/kW]**				**FOM [e/kW]**				**VOM [e/MWh]**
	2020	2030	2040	2050	2020	2030	2040	2050	
Nuclear	4.3	4.3	4.3	4.3	60	60	60	60	3.5
CCGT	0.9	0.9	0.9	0.9	25.7	25.7	25.7	25.7	2.4
OCGT	0.6	06	0.6	0.6	12	12	12	12	2.4
Wind onshore	1.5	1.3	1.1	0.9	35.7	32.7	30.3	27.8	0
Wind offshore	2.4	1.5	1.2	1.1	98.0	62.0	52.4	47.1	0
Solar PV	2.2	0.9	0.8	0.7	16.6	7.1	6.2	5.4	0

Table A3
Technology-specific costs of the considered storage technologies used in this paper. Data is roughly based on [39].

Technology	**Investment (power) [e/kW]**				**Investment (energy) [e/kWh]**				**VOM [e=/MWh]**
	2020	2030	2040	2050	2020	2030	2040	2050	
Battery	42.6	25.3	16.2	10.4	440.6	224.1	143.4	91.8	0

Appendix B. Basis planning model

B1. Objective function

The objective function expresses a minimization of the discounted total system costs, which consists of investment costs, fixed operation and maintenance (FOM) costs, variable operation and maintenance (VOM) costs, generation costs, and load-shedding costs.

$$Min \sum_{y\in Y} DF_y \cdot \left(c_y^{inv} + c_y^{fom} + c_y^{vom} + c_y^{gen} + c_y^{ens}\right) \tag{B.1}$$

The investment costs are a function of the investment variables for generation and storage technologies. The investment costs take into account the salvage fraction (SF) of each investment, i.e., the fraction that reflects the remaining value of the investments at the end of the planning horizon. Furthermore, we assume that the investment decision occurs at the start of each investment period so that these costs need to be discounted from the start of the period, to the period's milestone year DF_y^{SM}.

$$c_y^{inv} = DF_y^{SM} \cdot \Bigg[\sum_{g\in G} \left(1 - SF_{g,y}\right) \cdot C_{g,y}^{INV} \cdot cap_{g,y} + \sum_{s\in S} \left(1 - SF_{s,y}\right) \cdot \left(C_{s,y}^{INV,E} \cdot cap_{s,y}^{E} + C_{s,y}^{INV,P} \cdot cap_{s,y}^{P}\right)\Bigg] \quad \forall y \in Y \tag{B.2}$$

A similar expression is obtained for the FOM costs. However, to obtain the total FOM costs of an investment decision the annual FOM cost is multiplied by the discounted number of operating years (N^{OY}).

$$c_y^{fom} = DF_y^{SM} \cdot \left(\sum_{g\in G} N_{g,y}^{OY} \cdot C_{g,y}^{FOM} \cdot cap_{g,y} + \sum_{s\in S} N_{s,y}^{OY} \cdot C_{s,y}^{FOM,P} \cdot cap_{s,y}^{P}\right) \quad \forall y \in Y \tag{B.3}$$

The operational costs include the VOM costs, the generation costs and the load-shedding costs. These costs are calculated based on the representative days used to represent a characteristic year. Each representative day is assumed to be repeated W_i times within a characteristic year, and as such, the operational costs associated with that representative day are multiplied by W_i. Furthermore, to obtain the total operational costs, the annual operational costs are multiplied by the discounted number of years in the associated investment period (N_y^{PY}).

$$c_y^{vom} = N_y^{PY} \cdot \sum_{i\in I} W_i \cdot \left(\sum_{t\in T}\sum_{g\in G}\sum_{v\in V} C_{g,v}^{VOM} \cdot gen_{g,v,y,i,t} \cdot \Delta + \sum_{t\in T}\sum_{s\in S}\sum_{v\in V} C_{s,v}^{VOM} \cdot ch_{s,v,y,i,t} \cdot \Delta\right) \quad \forall y \in Y \tag{B.4}$$

The generation costs contain both the fuel as well as the emission costs associated with generating electricity.

$$c_y^{gen} = N_y^{PY} \cdot \sum_{i \in I} W_i \left[\sum_{t \in T} \sum_{g \in G} \sum_{v \in V} \left(\frac{C_{g,y}^{FUEL}}{\eta_{g,v}} + \frac{C^{GHG} \cdot EF_g}{\eta_{g,v}} \right) \cdot gen_{g,v,y,i,t} \cdot \Delta \right] \quad \forall y \in Y \tag{B.5}$$

The load shedding cost are directly proportional to the assumed value of lost load (VOLL).

$$c_y^{ens} = N_y^{PY} \cdot \sum_{i \in I} W_i \cdot \left(\sum_{t \in T} VOLL \cdot ens_{y,i,t} \cdot \Delta \right) \quad \forall y \in Y \tag{B.6}$$

B1.1. Power balance constraint

Eq. (B.7) expresses the balance between supply and demand.

$$\sum_{g \in G} \sum_{v \in V} gen_{g,v,y,i,t} + \sum_{s \in S} \sum_{v \in V} dc_{s,v,y,i,t} + ens_{y,i,t} = D_{y,i,t} + \sum_{s \in S} \sum_{v \in V} ch_{s,v,y,i,t} \quad \forall y \in Y; i \in I; t \in T \tag{B.7}$$

B2. Capacity transfer constraints

To translate investment variables into available capacity variables, three sets of capacity transfer constraints are used, namely one for investments in generation technologies, one for investments is the energy components of storage technologies and one for investments is the power components of storage technologies. The available capacity for year y of an investment characterized by a technology g and a vintage year v is equal to the investment decision variable times a CPT-factor. The CPT-factor is a parameter that reflects the fraction of an investment that is still available in year y and depends on the lifetime and the lead-time of the investment.

$$cap_{g,v,y}^{av} = CPT_{g,v,y} \cdot cap_{g,v} \quad \forall g \in G^{new}; v \in V; y \in Y \tag{B.8}$$

$$cap_{s,v,y}^{av,E} = CPT_{s,v,y} \cdot cap_{s,v}^{E} \quad \forall s \in S; v \in V; y \in Y \tag{B.9}$$

$$cap_{s,v,y}^{av,P} = CPT_{s,v,y} \cdot cap_{s,v}^{P} \quad \forall s \in S; v \in V; y \in Y \tag{B.10}$$

B3. Generation constraints

Eqs. (B.11) and (B.12) reflect the generation limits for respectively, newly installed dispatchable capacities and dispatchable legacy capacity.

$$gen_{g,v,y,i,t} \leq cap_{g,v,y}^{av} \quad \forall g \in G_D^{new}; v \in V; y \in Y; i \in I; t \in T \tag{B.11}$$

$$gen_{g,v,y,i,t} \leq CAP_{g,v,y}^{LEG} \quad \forall g \in G_D^{leg}; v \in V; y \in Y; i \in I; t \in T \tag{B.12}$$

For iRES technologies, the generation limits for new investments and legacy capacity are reflected by Eqs. (B.13) and (B.14).

$$gen_{g,v,y,i,t} + curt_{g,v,y,i,t} = CF_{g,i,t} \cdot cap_{g,v,y}^{av} \quad \forall g \in G_R^{new}; v \in V; y \in Y; t \in T; i \in T \tag{B.13}$$

$$gen_{g,v,y,i,t} + curt_{g,v,y,i,t} = CF_{g,i,t} \cdot CAP_{g,v,y}^{LEG} \quad \forall g \in G_R^{leg}; v \in V; y \in Y; t \in T; i \in T \tag{B.14}$$

B4. Storage constraints

Eq. (B.15) relates the power capacity of a storage technology to the energy capacity via the E/P-ratio.

$$cap_{s,y}^{E} = EP_s \cdot cap_{s,y}^{P} \quad \forall s \in S; y \in Y \tag{B.15}$$

Eqs. (B.16) and (B.17) express that storage units cannot charge or discharge more than the installed power capacity.

$$ch_{s,v,y,i,t} \leq cap_{s,v,y}^{av,P} \quad \forall s \in S; v \in V; y \in Y \tag{B.16}$$

$$dc_{s,v,y,i,t} \leq cap_{s,v,y}^{av,P} \quad \forall s \in S; v \in V; y \in Y \tag{B.17}$$

The temporal structure reflected by the representative days poses a challenge regarding the modeling of arbitrage opportunities over longer periods. Since (i) storage technologies should be allowed to net charge in one representative day and net discharge in another, and (ii) each representative days is repeated W_i times, tracking the state-of-charge of storage units requires information regarding the sequence in which these repetitions occur. In this model, we assume that each representative day is repeated W_i times before moving on to the next representative day. Following this assumption, the energy levels at the start of each representative day can be linked through the repetition of the hourly charge and discharge patterns.

$$e_{s,v,y,i+1,t=1}^{f} = e_{s,v,y,i,t=1}^{f} + W_i \cdot \sum_{t \in T} \left(\sqrt{\eta_{s,v}} \cdot ch_{s,v,y,i,t} \cdot \Delta - \frac{dc_{s,v,y,i,t}}{\sqrt{\eta_{s,v}}} \cdot \Delta \right) \quad \forall s \in S; v \in V; y \in Y; i \in I \tag{B.18}$$

To make sure that the energy level does not exceed the installed energy capacity, we need to track the energy level for all representative days and their repetitions. However, given our assumption regarding the sequential repetition of each representative day, it is safe to state that if the energy level violates the capacity limits in any of the repetitions, this is certainly true for the first and/or last repetition. From this it can be inferred that if the energy level does not violate capacity limits in the first and last repetitions of the representative day, capacity limits are not violated in any of the repetitions. As such, it is sufficient to track the energy levels in the first and last repetitions, which is the purpose of Eqs. (B.19) to (B.21).

$$e_{s,v,y,i,t+1}^{f} = e_{s,v,y,i,t}^{f} + \sqrt{\eta_{s,v}} \cdot ch_{s,v,y,i,t} \cdot \Delta - \frac{dc_{s,v,y,i,t}}{\sqrt{\eta_{s,v}}} \cdot \Delta \quad \forall s \in S; v \in V; y \in Y; i \in I; t \in T \tag{B.19}$$

$$e_{s,v,y,i,t=1}^{l} = e_{s,v,y,i,t=1}^{f} + (W_i - 1) \cdot \sum_{t \in T} \left(\sqrt{\eta_{s,v}} \cdot ch_{s,v,y,i,t} \cdot \Delta - \frac{dc_{s,v,y,i,t}}{\sqrt{\eta_{s,v}}} \cdot \Delta \right) \quad \forall s \in S; v \in V; y \in Y; i \in I \tag{B.20}$$

$$e_{s,v,y,i,t+1}^{l} = e_{s,v,y,i,t}^{l} + \sqrt{\eta_{s,v}} \cdot ch_{s,v,y,i,t} \cdot \Delta - \frac{dc_{s,v,y,i,t}}{\sqrt{\eta_{s,v}}} \cdot \Delta \quad \forall s \in S; v \in V; y \in Y; i \in I; t \in T \tag{B.21}$$

Finally, Eq. (B.22) and (B.23) prevent the energy levels of the first and last repetitions to exceed the available energy capacity.

$$e_{s,v,y,i,t}^{f} \leq cap_{s,v,y}^{av,E} \quad \forall s \in S; v \in V; y \in Y \tag{B.22}$$

$$e_{s,v,y,i,t}^{l} \leq cap_{s,v,y}^{av,E} \quad \forall s \in S; v \in V; y \in Y \tag{B.23}$$

This representation is likely to overly constrain the storage technologies' energy capacity when a low number of representative days are used [40].

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